CentOS 7.4 Base Installation and Lab Config

Base Cent OS 7.4 VM for use with BIND (DNS) server or other ArcSight components (such as ArcMC, Logger, etc.)

- **BIND installation (for 172.16.100.x) lab network**
- **ArcMC installation (ArcMC 2.81)**
- **Logger management by ArcMC (Logger 6.61)**

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CentOS 7.4 Installation

Accessing Vanilla CentOS ("Minimal" and Full DVD Installs)

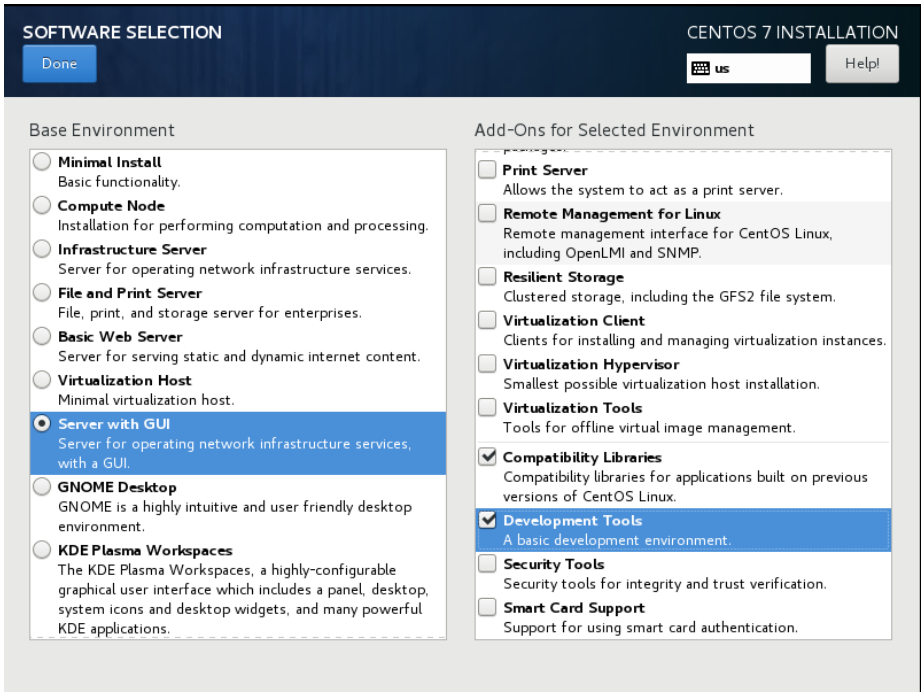
- CentOS 7.4 available from:
  - http://archive.kernel.org/centos-vault/7.4.1708/isos/x86_64/
    - CentOS-7-x86_64-Minimal-1708.iso
    - CentOS-7-x86_64-DVD-1708.iso (alternate option for full install)

- For other versions of CentOS, can access main vault dir:
  - http://archive.kernel.org/centos-vault

Base Install of OS (CentOS 7.4) <VM Example – Example "Starting Point">

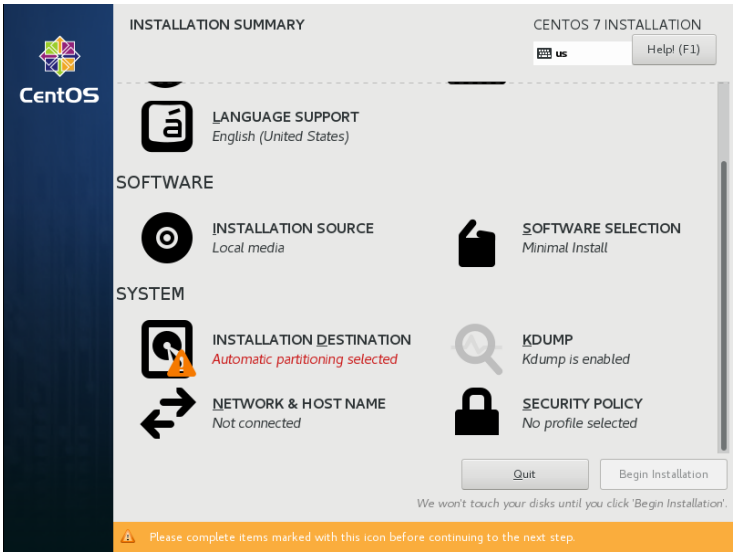
- Start VMware (used VMware Workstation 12 Pro v. 12.5.9 build-7535481)
- Select File > New Virtual Machine
- Leave Typical, click Next
- Select "Installer disc image file (iso):
  - For CentOS 7.4 full install, browse to: CentOS-7-x86_64-DVD-1708.iso
- Select ISO and click Open
- Click Next
- Set virtual machine name:
  - For a vanilla starting point VM: `macvanlinux.example.com`
- Set Location to where VM will be installed
  - For example: `C:\macgyver_vmware\macvanlinux.example.com`
- Click Next
- Set maximum disk size to 100G
  - For virtual systems, leave default of "Split virtual disk into multiple files"
- Click Next
- Click Customize Hardware
- Set Memory to 8192 (8G)
- Set # of processors to 2
- Set # of cores to processor to 4 (8 total processor cores)
  - Match settings to hardware availability
- Left virtualization engine default on "Automatic"
- Set Network Adapter
  - If using a VM IP *different* than the network to which VM system connected, use NAT
    - For example, setting up VM IP as 172.16.100.x (but your primary system network is 192.168.0.x)
  - If using a VM IP in range the same as system on which VM running, use Bridged
    - For example, setting up VM IP as 192.168.0.x (and your primary system network is 192.168.0.x)
- Remove the following devices (single click on each, and select Remove)
  - Sound Card
  - Printer
- Click Close
- At Summary window (Ready to Create Virtual Machine), click Finish
  - OS (CentOS or Red Hat) boots up in VM

- After reboot, pick option to Install CentOS
    - Press <Enter> to begin the installation process
        - Defragementation check may automatically show
    - After initial setup checks, the “Welcome to CentOS 7 window displays”
    - Leave default of English, and click Continue
    - On the Installation Summary window:
        - Click on Software Selection
            - For Base Environment, select:
                - Server with GUI
            - For Add-ons, select:
                - Compatibility Libraries
                - Development Tools

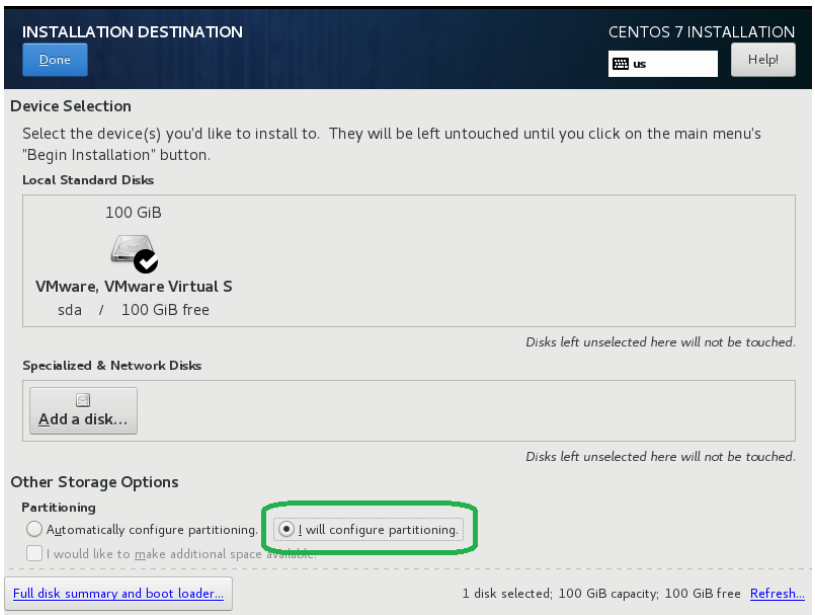


            - Click Done

- Click on Installation Destination to set disk partitions and partition types

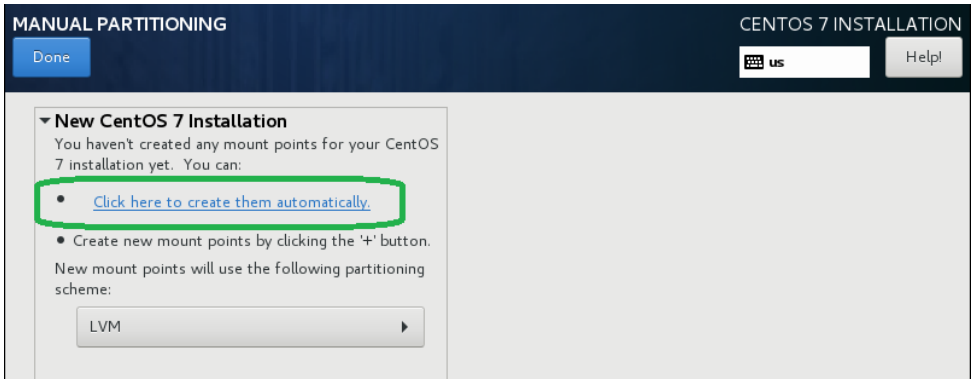


o On Installation Destination page, set Partitioning option to “I will configure partitioning.”

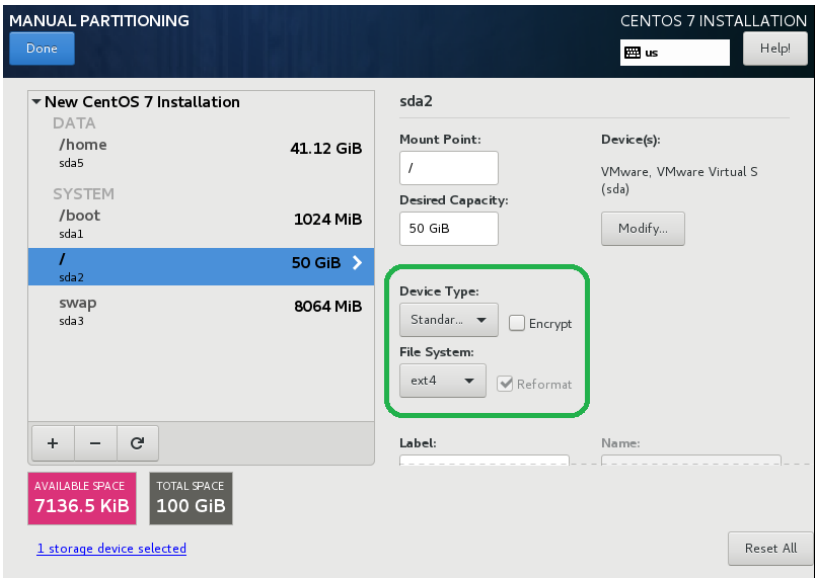


o Click Done

- To create mount points by default, click on “Click here to create them automatically”



- Click on each of the following partitions (/home, /boot, and /), and set:
  - Device Type = Standard
  - File System: ext4



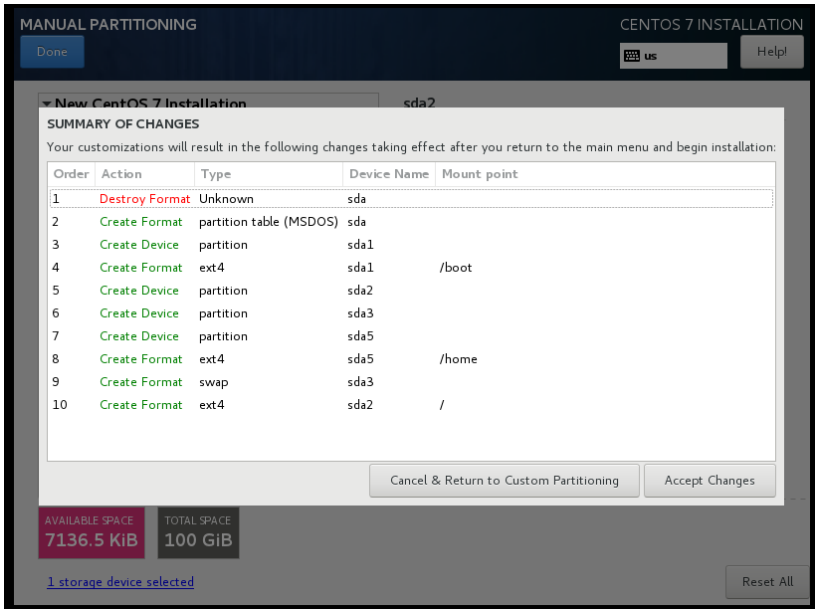
  - Leave swap set to “swap”

- Set partition sizing:
  - /home: 1 GiB
  - /boot: <leave default of 1024>
  - /: 90 GiB
  - swap: <leave default>

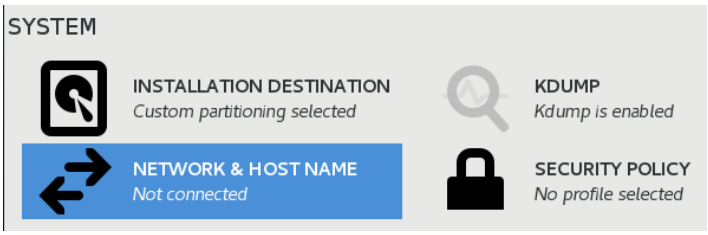
  This will leave a small amount of disk unallocated, which is fine

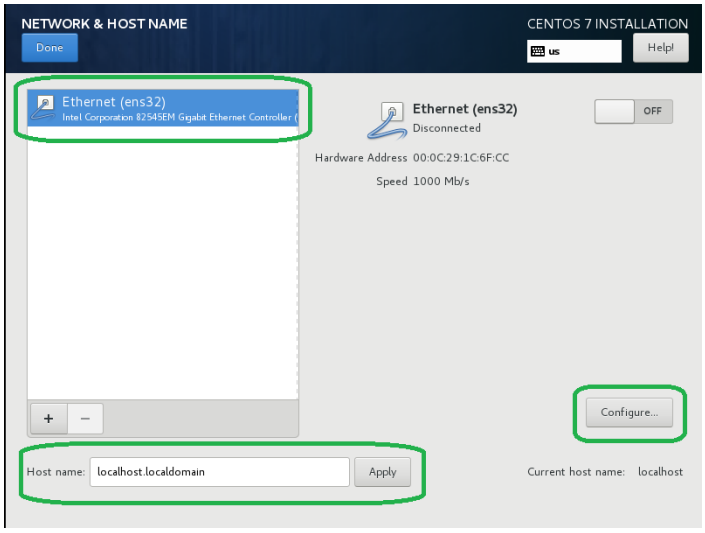
- Click Done

- When prompted with a summary of changes, click “Accept Changes”



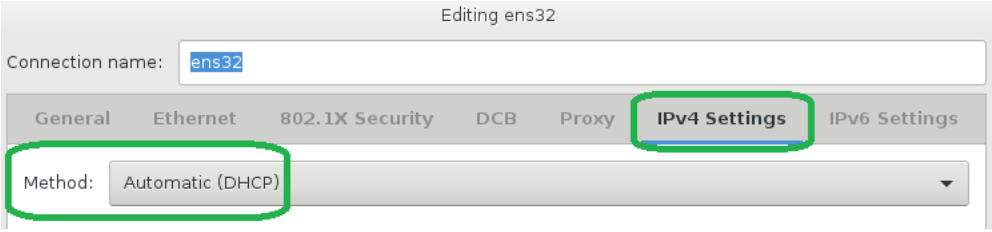
- Click Network & Host Name to set the networking-related values for the VM



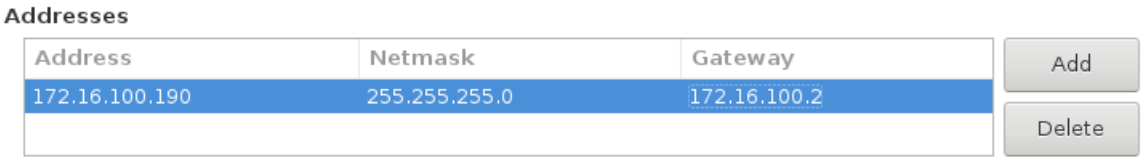


- Click in the Host name field, and set the hostname for the CentOS instance:
  - For example: macvanlinux (MacGyver vanilla Linux)
  - Click Apply
  - The “Current host name” label should update with the host name that was set

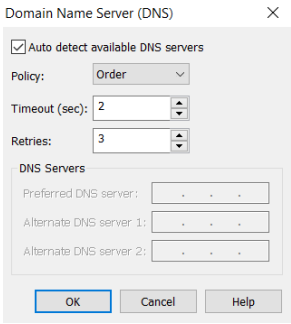
- Click on the Ethernet controller, and click the Configure button
  - Click on the IPv4 Settings tab



  - Change Method from “Automatic” to “Manual”
  - In the Addresses section, click Add
    - Fill in IP, Netmask, and Gateway and press <Enter>

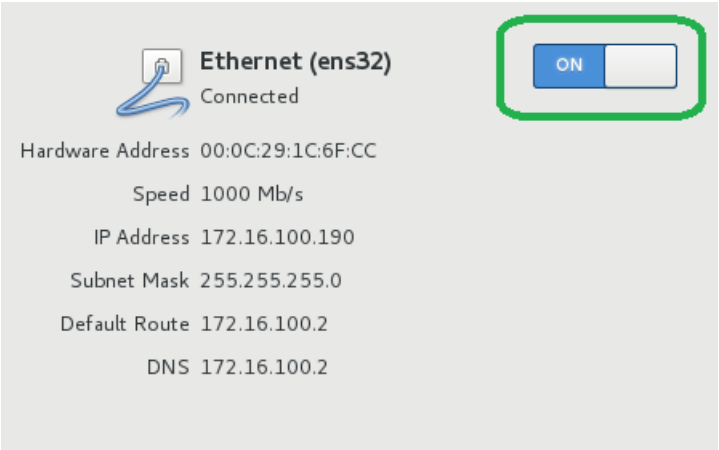


  - In the DNS servers section:
    - For use with EB, in this example, *set DNS to 172.16.100.2*
      - Within VM in NAT mode:
        - “The NAT device is a DNS proxy. It forwards DNS requests from the virtual machines to a DNS server that the host knows. Responses return to the NAT device, which then forwards them to the virtual machines.”
        - By default, in the NAT settings in VMware, DNS is set as follows (auto detection):



  - In the Search domains field, type:
    - `example.com`

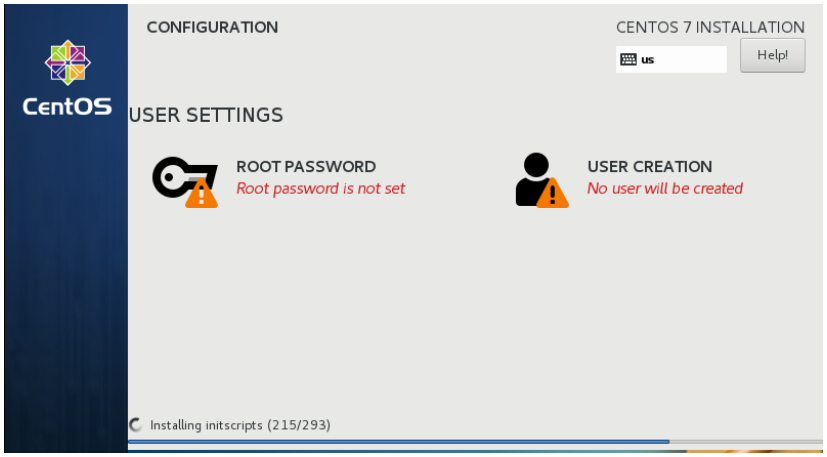
- Click Save
- Slide the Ethernet bar in upper-right to “On”, and verify settings



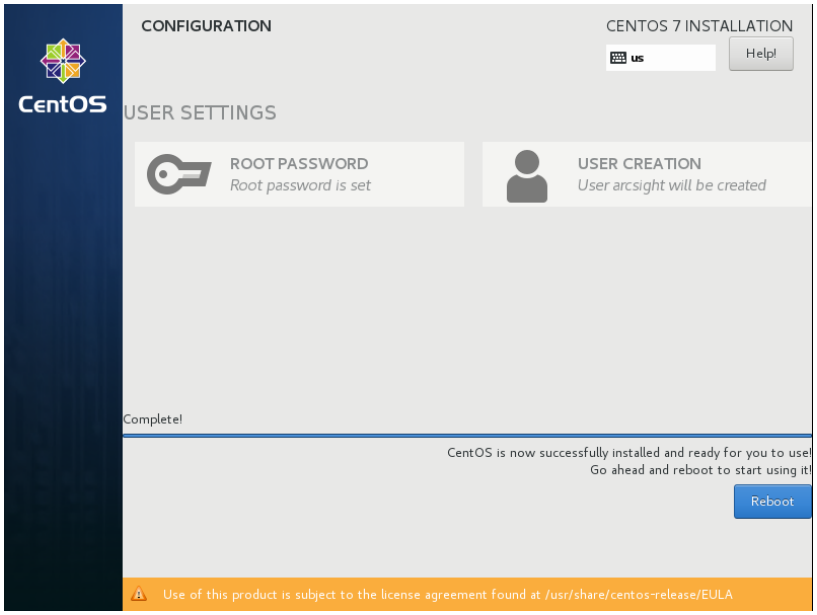
- Click Done

- Click Begin Installation
  - With CentOS 7.4 DVD1 (as of 8/16/18):
    - With “Server with GUI” option selected, ~1300 packages installed by default
      - If “Development and Creative Workstation” option was selected, ~1500 packages installed
      - If only minimal install used, only ~300 packages installed by default

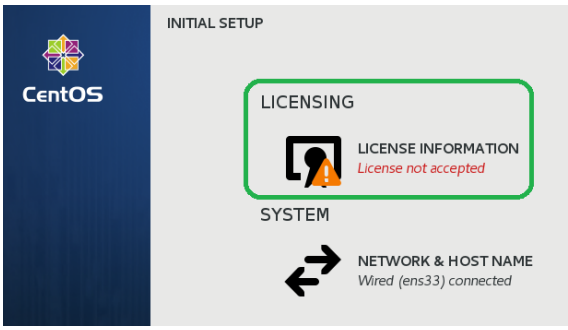
- While OS installing:
  - Set the root password
  - Create a default non-root user, such as “arcsight” (and set password for that user)
  - Click on each option and follow prompts
  - Click Done after each step complete (click Done twice if password weak)



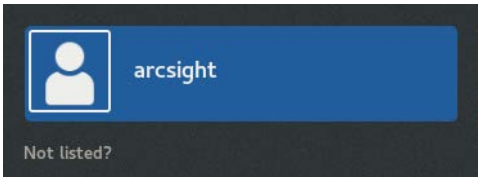
- When installation is complete, click Reboot

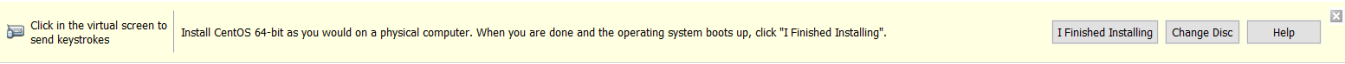


- After first reboot, click on LICENSE INFORMATION to accept license terms



  - Click I accept license agreement (bottom left of page)
  - Click Done
  - Click FINISH CONFIGURATION
- After OS login displays, click “I Finished Installing” (at bottom of VM window)





- SSH logon as root after reboot completes
  - Verify disk partitions and types by typing:

    lsblk –f

```
[root@macvanlinux ~]# lsblk -f
NAME            FSTYPE      LABEL UUID                                   MOUNTPOINT
sda
├─sda1          ext4              7b8c7749-18d6-4b78-9a52-cfbaab968b22   /boot
├─sda2          ext4              783f7183-9911-435d-85e9-c6f887319700   /
├─sda3          LVM2_member       zKNBbb-Uiq0-Mm2Q-RpYx-0Cna-j5VZ-5jnEx2
│ └─centos-swap swap              8e4fae30-5c4e-45f7-8c93-ec53d7154e23   [SWAP]
├─sda4
└─sda5          ext4              1943765b-3e8b-4928-8ea7-081a68444eb4   /home
```

  - Verify disk sizing by typing:

    df -k

```
[root@macvanlinux ~]# df -k
Filesystem     1K-blocks    Used Available Use% Mounted on
/dev/sda2       92759720 3873860  84150884   5% /
devtmpfs         3981208       0   3981208   0% /dev
tmpfs            3997036       0   3997036   0% /dev/shm
tmpfs            3997036    9176   3987860   1% /run
tmpfs            3997036       0   3997036   0% /sys/fs/cgroup
/dev/sda5         999320    2592    927916   1% /home
/dev/sda1         999320  152936    777572  17% /boot
tmpfs             799408       8    799400   1% /run/user/42
tmpfs             799408       0    799408   0% /run/user/0
```

- Add default entry to /etc/hosts:

```
172.16.100.95    macvanlinux.example.com
```

- Update process limits by typing:

  vi /etc/security/limits.d/20-nproc.conf

  Comment out the default soft nproc entries (start each line with #) and add these entries

```
* soft nproc 10240
* hard nproc 10240
* soft nofile 65536
* hard nofile 65536
* soft core unlimited
* hard core unlimited
```

- Verify system can ping an external address (requires system on which VM being installed has internet access):

  ping www.cnn.com

  - Verify name resolves, and external site reached
    - In this setup so far, the VMware host makes use of the local system (for example, local laptop) to resolve to external DNS.
    - External name resolution needed later for yum-related steps

- <optional> Can also add entry for the hostname to your local system hosts file
    - For example, so you can easily ssh by hostname to the system
    - On Windows 10, for example:

```
C:\Windows\System32\drivers\etc\hosts
```

Post-Base OS Installation Steps (Includes prep for EB)

- Update the OS kernel

```
yum update -y kernel
```

    i. The kernel must be 3.10.0-693.21.1.el7.x86_64 or higher for Event Broker

- Install (or upgrade) additional pre-req packages and dependencies:

```
yum install -y bind-utils unzip nfs-utils libseccomp libtool-ltdl
chrony java-1.8.0-openjdk-headless lsof net-tools conntrack-tools
httpd-tools perl tcpdump
```

    i. java-1.8.0-openjdk-headless is required on EB Master nodes but is not required on the EB Worker nodes
    ii. bind-utils is the package that includes the nslookup capability
    iii. perl is required for ArcMC agent to install
    iv. tcpdump can assist with network-based troubleshooting

- Configure chrony by typing:
    a. Chrony synchronizes the system clock with external time sources, using the NTP protocol, whenever network access allows it to do so

```
systemctl start chronyd
systemctl enable chronyd
chronyc tracking
```

- Set run level for system to start in text mode vs. GUI mode
    a. # To view current default target, run:

```
systemctl get-default
```

    Example:

```
[root@macvanlinux ~]# systemctl get-default
graphical.target
```

    b. # To set to non-graphical target (quicker bootup), run:

```
systemctl set-default multi-user.target
```

    Example:

```
[root@macvanlinux ~]# systemctl set-default multi-user.target
Removed symlink /etc/systemd/system/default.target.
Created symlink from /etc/systemd/system/default.target to
/usr/lib/systemd/system/multi-user.target.
```

- Reboot VM
  a. Can type either of the following, for example:
     `reboot`
     or
     `shutdown –h now` (and then re-power on VM manually)

- Log back on as root
- Check that `20-nproc.conf` file was properly modified
  a. Type:
     `ulimit -a`
     i. Verify results match following entries

```
core file size          (blocks, -c) unlimited
data seg size           (kbytes, -d) unlimited
scheduling priority             (-e) 0
file size               (blocks, -f) unlimited
pending signals                 (-i) 31091
max locked memory       (kbytes, -l) 64
max memory size         (kbytes, -m) unlimited
open files                      (-n) 65536
pipe size            (512 bytes, -p) 8
POSIX message queues     (bytes, -q) 819200
real-time priority              (-r) 0
stack size              (kbytes, -s) 8192
cpu time               (seconds, -t) unlimited
max user processes              (-u) 10240
virtual memory          (kbytes, -v) unlimited
file locks                      (-x) unlimited
```

Create Vanilla Linux Snapshot for use as Starting Point in ArcSight Installations

- Shut down VM
- Add a description of the VM, including IP, and user details to the "Description" section
  - For example:

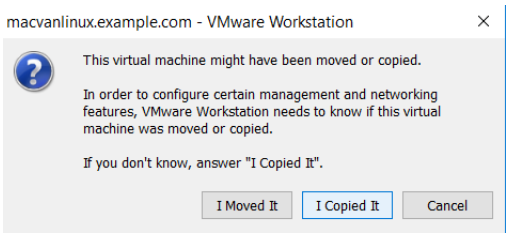
```
CentOS 7.4
Base: Server with GUI

Host: macvanlinux
FQDN: macvanlinux.example.com
IP: 172.16.100.95

root/root
arcsight/root
```

- Take a snapshot

DNS (BIND) Server Installation

Copying Vanilla OS VM to Use for BIND Server

- Copied “macvanlinux.example.com” VM default image folder to “dns.example.com”
- From VMware, open the VMX file in that folder (but don’t start VM yet)
- In VMware, update the description for the VM to planned target.
    - For example:
        - Host: dns
        - FQDN: dns.example.com
        - IP: 172.16.100.53
- In local Windows system, update hosts file with these new details for the target host
- In VMware, right-click on the tab for the VM, and click Settings
    - Click Options
    - Change virtual machine name to updated name (e.g.: dns.example.com)
    - Click Ok
- Update hardware settings
    - Reduce the Memory setting to:
        - 4096 MB
    - Reduce the Processors setting to:
        - 1 Processor
        - 2 cores per processor
- Power on the VM
    - When prompted, click “I Copied It”



- Log on as root
- Update /etc/hosts
    - Update entry:

```
172.16.100.53      dns.example.com
```

    - Add any planned entries:

```
172.16.100.242 arcmcvan.example.com
172.16.100.191 eb1.example.com
172.16.100.100 logger.example.com
```

- Use nmtui:
  - Set system hostname to “dns”
  - Edit default connection (in my case, “ens33”):
    - Change IP address to: 172.16.100.53
    - Verify Gateway to: 172.16.100.2
    - Update DNS servers to point to:
      - First: 172.16.100.53
      - Second: 172.16.100.2
    - Verify search domain set to: example.com

- Reboot VM

Disable Firewall

Before installing or upgrading software ArcMC (in a lab or test environment), disable the firewall to ensure no network communication issues.

- As root, type the following commands:

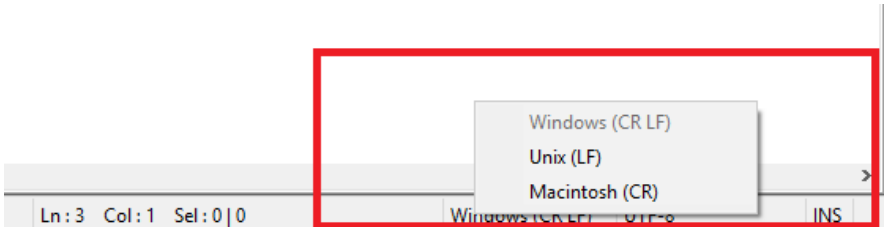
```
systemctl status firewalld
systemctl stop firewalld
systemctl disable firewalld
systemctl status firewalld
```

Install BIND

1. On DNS server, as root, install BIND by typing:

```
yum -y install bind bind-utils
```

2. Create four files with the names shown below using Notepad++. *(In Appendix of this guide, are contents you can copy and paste into each of these files.)*
   a. Modify the files based on your IP address space
      i. The forward and reverse zones are based on the 172.16.100.x Class C private address space. You’ll need to modify them based on the IP address space and hostnames you are using.
   b. NOTE: Make sure you save the files as Unix (LF).
      i. If using Notepad++, option to save in the “Unix (LF)” format on bottom-right of Notepad++ window



   c. example.com and 100.16.172.in-addr.arpa need to have a blank line at the end of the file.
   d. Reboot VM

e. Backup the existing /etc/named.conf file

```
mv /etc/named.conf /etc/named.conf.bak
```

f. Create /etc/named/zones directory

```
mkdir /etc/named/zones
```

g. Copy the BIND files to the following directories.
   i. /etc
      1. named.conf
   ii. /etc/named
      1. named.conf.local
   iii. /etc/named/zones
      1. example.com
      2. 100.16.172.in-addr.arpa

3. From command line, run following commands and verify the output:

```
named-checkconf
```

   i. If there are no errors then there should be no output from this command.

```
named-checkzone example.com /etc/named/zones/example.com
```

   ii. Output: `zone example.com/IN: loaded serial 0 OK`

```
named-checkzone 100.16.172.in-addr.arpa
/etc/named/zones/100.16.172.in-addr.arpa
```

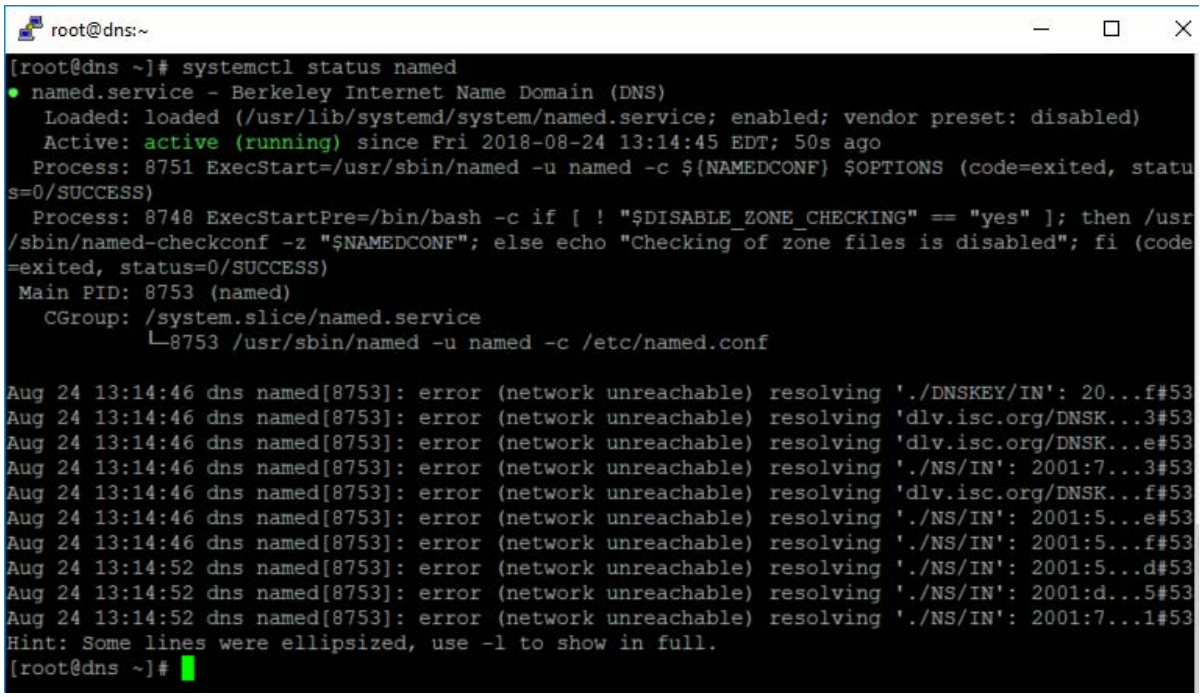
   iii. Output: `zone 100.16.172.in-addr.arpa/IN: loaded serial 0 OK`

4. Enable BIND on boot and start it

```
systemctl enable named
systemctl start named
```

5. Verify that the BIND service is running

```
systemctl status named
```



```
root@dns:~
[root@dns ~]# systemctl status named
● named.service - Berkeley Internet Name Domain (DNS)
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; vendor preset: disabled)
   Active: active (running) since Fri 2018-08-24 13:14:45 EDT; 50s ago
  Process: 8751 ExecStart=/usr/sbin/named -u named -c ${NAMEDCONF} $OPTIONS (code=exited, statu
s=0/SUCCESS)
  Process: 8748 ExecStartPre=/bin/bash -c if [ ! "$DISABLE_ZONE_CHECKING" == "yes" ]; then /usr
/sbin/named-checkconf -z "$NAMEDCONF"; else echo "Checking of zone files is disabled"; fi (code
=exited, status=0/SUCCESS)
 Main PID: 8753 (named)
   CGroup: /system.slice/named.service
           └─8753 /usr/sbin/named -u named -c /etc/named.conf

Aug 24 13:14:46 dns named[8753]: error (network unreachable) resolving './DNSKEY/IN': 20...f#53
Aug 24 13:14:46 dns named[8753]: error (network unreachable) resolving 'dlv.isc.org/DNSK...3#53
Aug 24 13:14:46 dns named[8753]: error (network unreachable) resolving 'dlv.isc.org/DNSK...e#53
Aug 24 13:14:46 dns named[8753]: error (network unreachable) resolving './NS/IN': 2001:7...3#53
Aug 24 13:14:46 dns named[8753]: error (network unreachable) resolving 'dlv.isc.org/DNSK...f#53
Aug 24 13:14:46 dns named[8753]: error (network unreachable) resolving './NS/IN': 2001:5...e#53
Aug 24 13:14:46 dns named[8753]: error (network unreachable) resolving './NS/IN': 2001:5...f#53
Aug 24 13:14:52 dns named[8753]: error (network unreachable) resolving './NS/IN': 2001:5...d#53
Aug 24 13:14:52 dns named[8753]: error (network unreachable) resolving './NS/IN': 2001:d...5#53
Aug 24 13:14:52 dns named[8753]: error (network unreachable) resolving './NS/IN': 2001:7...1#53
Hint: Some lines were ellipsized, use -l to show in full.
[root@dns ~]#
```

6. Verify DNS returns values for nslookups:
   a. nslookup dns (returns local DNS server)

```
[root@dns etc]# nslookup dns
Server:         172.16.100.53
Address:        172.16.100.53#53

Name:   dns.example.com
Address: 172.16.100.53
```

   b. nslookup www.cnn.com (returns external DNS results from VM .2 interface)

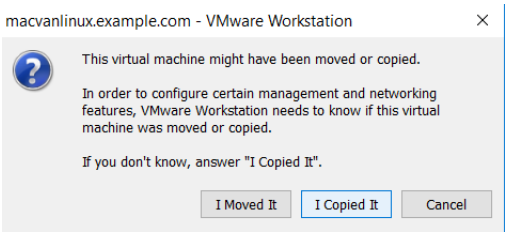
```
[root@dns etc]# nslookup www.cnn.com
Server:         172.16.100.2
Address:        172.16.100.2#53

Non-authoritative answer:
www.cnn.com     canonical name = turner-tls.map.fastly.net.
Name:   turner-tls.map.fastly.net
Address: 151.101.49.67
```

7. Shut down DNS VM, and save a snapshot

Vanilla ArcMC Installation

New ArcMC Install Based on vanilla CentOS VM

- Copied “macvanlinux.example.com” VM default image folder to “arcmcvan.example.com”
- From VMware, open the VMX file in that folder (but don’t start VM yet)
- In VMware, update the description for the VM to planned target.
    - For example:
        - Host: arcmcvan
        - IP: 172.16.100.242
        - FQDN: arcmcvan.example.com
- In VMware, right-click on the tab for the VM, and click Settings
    - Click Options
    - Change virtual machine name to updated name (e.g.: arcmcvan2.example.com)
    - Click Ok
- In local Windows system, update hosts file with these new details for the target host
- Power on the VM
    - When prompted, click “I Copied It”



- Log on as root
- Update /etc/hosts
    - Update entry:

        172.16.100.242 arcmcvan.example.com

        NOTE: Apache service on ArcMC will not run unless ArcMC can resolve server host name on which ArcMC installed (either through /etc/hosts or DNS)

    - Add any planned entries:

        172.16.100.53 dns.example.com
        172.16.100.191 eb1.example.com
        172.16.100.100 logger.example.com

- Use nmtui:
    - Set system hostname to “arcmcvan”
    - Edit default connection (in my case, “ens33”):
        - Change IP address to: 172.16.100.242
        - Verify Gateway to: 172.16.100.2
        - Update DNS servesr to point to:
            - First: 172.16.100.53
            - Second: 172.16.100.2
        - Verify search domain set to: example.com

- Reboot VM

Disable Firewall

Before installing or upgrading software ArcMC (in a lab or test environment), disable the firewall to ensure no network communication issues.

- As root, type the following commands:

```
systemctl status firewalld
systemctl stop firewalld
systemctl disable firewalld
systemctl status firewalld
```

Post-Base OS Installation Steps (Prep for ArcMC)

Verify Perl installed

- Verify perl installed (required for local ArcMC agent) – re-verifies connectivity to mirror, and if any updates required:

```
yum install perl
```

Ensure Host Resolvable

- For the Apache web process to start, the Software ArcSight Management Center hostname must be resolvable. Add the ArcMC hostname to either /etc/hosts or DNS

```
ping arcmcvan
```

Update logind.conf on ArcMC

Before installing or upgrading software ArcMC on (CentOS and) Red Hat Enterprise Linux (RHEL) 7.X, you must modify the inter-process communication (IPC) setting of the logind.conf file.

- As root, navigate to the /etc/systemd directory
- Open the logind.conf file for editing

- Find the RemoveIPC line
    - RemoveIPC should be active and set to no.
        - Remove the # sign if it is there
        - Change the yes to no if needed
    - The correct entry is: RemoveIPC=no (line not commented out)
- Save the file
- From the /etc/systemd directory, enter the following command to restart the systemd-logind service and put the change into effect:

```
systemctl restart systemd-logind.service
```

Install ArcMC (arcmcvan)

- Logon as root
- Copy ArcMC bin file to /tmp on ArcMC server
- Follow default ArcMC installation instructions from ArcMC:
    - Copy ArcMC bin to ArcMC server (for example, to /tmp)
    - Type:

```
chmod +x ArcSight-ArcMC-2.8.1.2143.0.bin
```

    - Type:

```
./ArcSight-ArcMC-2.8.1.2143.0.bin -i console
```

- Choose default install options:
    - For this guide, install path:

```
/opt/arcsight
```

    - Accept all defaults for installation
        - For this guide, when prompted for non-root user, using “arcsight”
        - Skipped license file (using instant-on license by default to start)
- After all installation steps complete, check service status with the following command:

```
/opt/arcsight/current/arcsight/arcmc/bin/arcmcd status
```

Example output before first reboot:

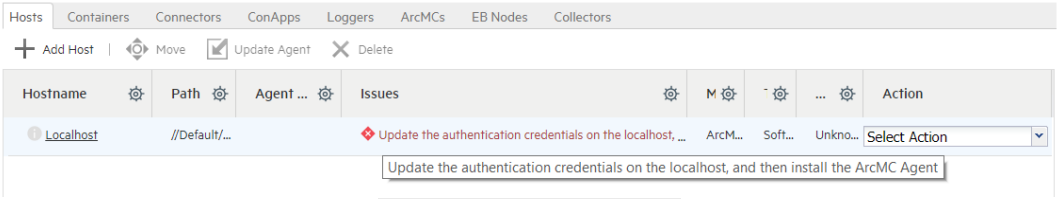
```
Process 'apache'                    running
Process 'aps'                       running
Process 'postgresql'                running
Process 'web'                       running
```

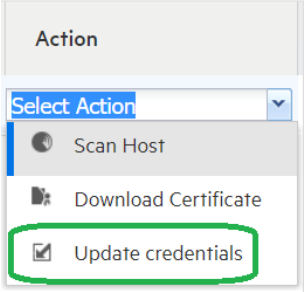
- Reboot ArcMC VM
- Log on to ArcMC UI
    - When prompted, re-set password
- Click Node Management > View All Nodes
- Click on the Default location

- In the Hosts tab, for Localhost row, in the Action dropdown, if credential error displays:

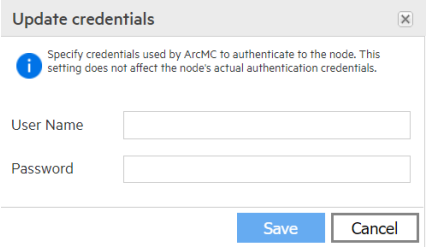
  <u>NOTE: If errors do not display, can skip this section.</u>

  - Click Update credentials (because default password for admin changed at first login)

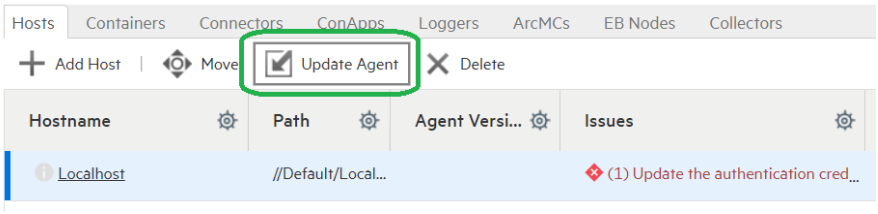




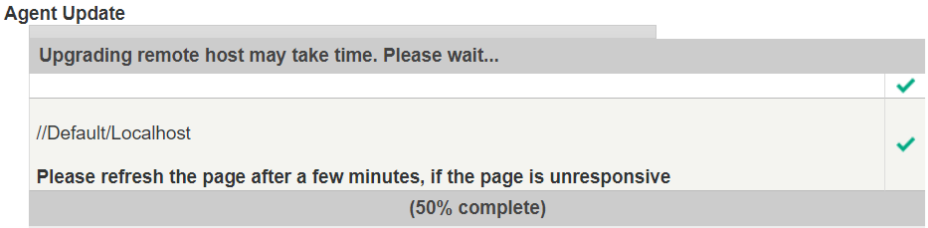
For user admin, update password, then click Save:



- Click on the Localhost row, and select Update Agent



- Click Next



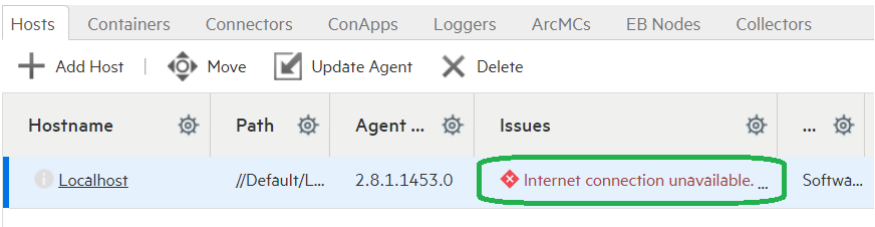
- From command line, re-check service status:

```
/opt/arcsight/current/arcsight/arcmc/bin/arcmcd status
```

Example output after ArcMC agent updated:

```
Process 'apache'                    running
Process 'aps'                       running
Process 'arcmcagent'                running
Process 'postgresql'                running
Process 'web'                       running
```

After a few minutes, ArcMC will update the Localhost status in the UI. (In this example, ArcMC system not on network with internet access):



- Delete the ArcMC bin file from the /tmp dir (or whichever dir bin originally copied into) post installation
  - Done to save space on the VM snapshot
- Shutdown ArcMC and take a snapshot

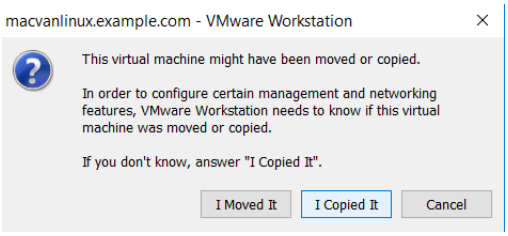
EB Server Installation and Management by ArcMC

Companion Notes to Micro Focus Enablement Ground-up Build Guide

- For this guide, I am using the 172.16.100.x network for VMs (including DNS at 172.16.100.53)
    - For EB:
        - Hostname: eb1
        - IP: 172.16.100.191
        - FQDN (in /etc/hosts) as: eb1.example.com
    - For suite (cluster) and ArcSight Installer user / password:
        - admin / Arcs1ght!

New EB Install Based on vanilla CentOS VM

- Copied “macvanlinux.example.com” VM default image folder to “eb1.example.com”
- From VMware, open the VMX file in that folder (but don’t start VM yet)
- In VMware, update the description for the VM to planned target.
    - For example:
        - Host: eb1
        - IP: 172.16.100.191
        - FQDN: eb1.example.com
- In VMware, right-click on the tab for the VM, and click Settings
    - Click Options
    - Change virtual machine name to updated name (e.g.: eb1.example.com)
    - Click Ok
- In local Windows system, update hosts file with these new details for the target host
- Power on the VM
    - When prompted, click “I Copied It”



- Log on as root

- Update /etc/hosts

  Update hosts file exactly as shown (no short host name in list).
  On install, Event Broker uses hostname –f to verify FQDN, and if short host-name first in list, that is the value that will be returned (regardless of DNS).

  - Update entry:

    ```
    172.16.100.242 eb1.example.com
    ```

  - Add any planned entries:

    ```
    172.16.100.53     dns.example.com
    172.16.100.242 arcmcvan.example.com
    172.16.100.100 logger.example.com
    ```

- Use nmtui:
  - Set system hostname to "eb1"
  - Edit default connection (in my case, "ens33"):
    - Change IP address to: 172.16.100.191
    - Verify Gateway to: 172.16.100.2
    - Update DNS servesr to point to:
      - First: 172.16.100.53
      - Second: 172.16.100.2
    - Verify search domain set to: example.com

- Reboot VM
- Repeat following steps (done in earlier Vanilla OS steps) in case there are any updates from time the base vanilla OS was built:

  - Update the OS kernel

    ```
    yum update -y kernel
    ```

    - The kernel must be 3.10.0-693.21.1.el7.x86_64 or higher for Event Broker

  - Install (or upgrade) additional pre-req packages and dependencies:

    ```
    yum install -y bind-utils unzip nfs-utils libseccomp
    libtool-ltdl chrony java-1.8.0-openjdk-headless lsof net-
    tools conntrack-tools httpd-tools perl tcpdump
    ```

    - java-1.8.0-openjdk-headless is required on EB Master nodes but is not required on the EB Worker nodes
    - bind-utils is the package that includes the nslookup capability
    - perl is required for ArcMC agent to install
    - tcpdump can assist with network-based troubleshooting

- Boot DNS server (172.16.100.53)
- Shut down VM and take snapshot ("ready to install" stage)
- Follow steps outlined in Micro Focus' Technical Enablement ground-up EB Guide
- After ArcSight Installer installed, and EB pre-deploy ready, can check ArcSight Installer pods
  - Wait for all pods to be in "running" state before moving to EB deployment

    ```
    watch -n 15 kubectl get pods --all-namespaces
    ```

EB Deployment Notes

- Deploy Event Broker:
  - Browse to ArcSight Installer:

    https://eb1.example.com:5443/

  - Click Node Management
    - Verify status shows as ready (green check mark)

	Address	Role	Memory (GB)	CPU (cores)	Labels			Created On	Ready
	eb1.example.com [172.16.100.191]	MASTER	8 GB	8	zk:yes ×	kafka:yes ×	+	Sep 20, 2018 2:58:48 pm	✓

  - Click Deployment
    - Click Deploy in ArcSight Event Broker row

Deployment

	Name	Version	Details	Status	
	ArcSight Event Broker				DEPLOY

    - Choose version (2.21), and click Deploy

Deploy ArcSight Event Broker ×

Choose a version

◉ 2.21

DEPLOY

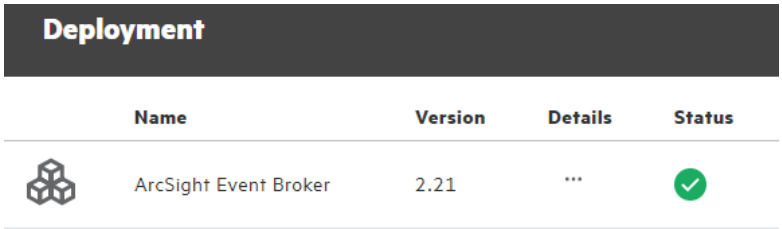
    - From a terminal window, can run the following command on EB1 to watch new Event Broker pods being deployed

```
watch -n 15 kubectl get pods --all-namespaces
```

New pods will include the following:

```
eventbroker1   eb-kafka-0                              1/1     Running   0          4m
eventbroker1   eb-kafka-manager-8ff559cb8-6trhs        1/1     Running   0          4m
eventbroker1   eb-routing-processor-0                  1/1     Running   0          4m
eventbroker1   eb-schemaregistry-6b77657b5f-mh56c      1/1     Running   0          4m
eventbroker1   eb-web-service-94fc85fb-vswgx           2/2     Running   0          4m
eventbroker1   eb-zookeeper-0                          1/1     Running   0          4m
eventbroker1   suite-reconf-pod-eventbroker-t5ghx      2/2     Running   0          4m
```

- Once Event Broker has been deployed, a green status check should also display in the installer



EB Deployment Monitoring (Tunnel to EB)

- Can use MobaTerm to set up a tunnel to connect to EB and view the status (from Micro Focus Technical Enablement)

  When Event Broker is finally installed, the port for the web interface is bound to a Docker interface that is not exposed to the network. From the console of the Event Broker node, you can connect locally to that IP address and port if you have X Windows and a browser installed on the node. But we did a minimal CentOS install, so the appropriate packages are not installed.

  The best option is to use an SSH tunnel to tunnel the traffic through the Event Broker node to the web interface. First, we need to find out the IP address that the Event Broker web interface is listening on. The port is going to be 9000 and we can find the IP address by running the following command and identifying the IP that 9000 is bound to. In this case, it is 172.30.78.65.

  Remember, this is not exposed to the network.

  NOTE: This IP address may change if Event Broker is undeployed or restarted.

  - Logon to EB server as root, and type following command:

    ```
    kubectl get service -n eventbroker1 | grep 9000
    ```

    Example response:

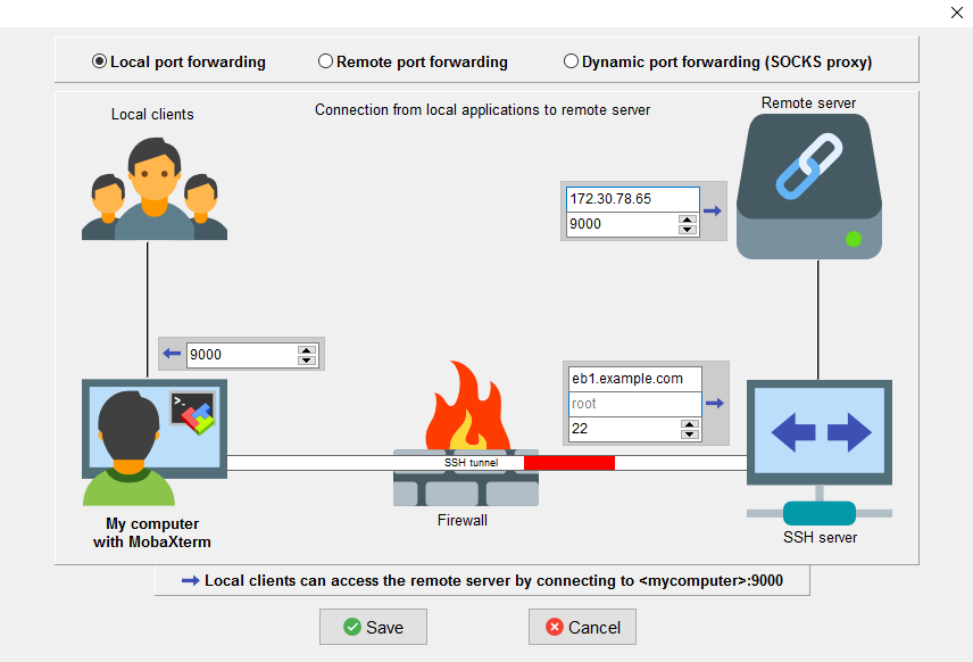
    ```
    eb-kafkamgr-svc  ClusterIP  172.30.78.65  <none>  9000/TCP
       3m
    ```

  I'm using MobaXterm (https://mobaxterm.mobatek.net/) and the instructions to setup SSH tunnelling are here: https://blog.mobatek.net/post/ssh-tunnels-and-port-forwarding/ .

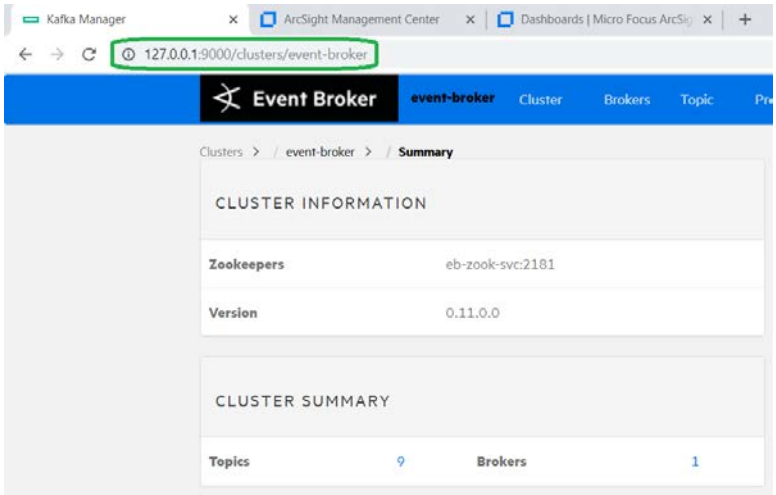
    - Once I've setup the SSH tunnel, when I start it, I can use my local browser and connect to http://127.0.0.1:9000/ to connect to the Event Broker web interface.

  Two things to note, this connection is not https and there is no authentication. We're tunnelling the insecure traffic through an SSH session, so that risk is partially mitigated, but there is no

authentication. This is why the Event Broker web interface is not publically exposed. Below are screenshots of the SSH tunnel.



Following screen is from the Event Broker after tunnel established:



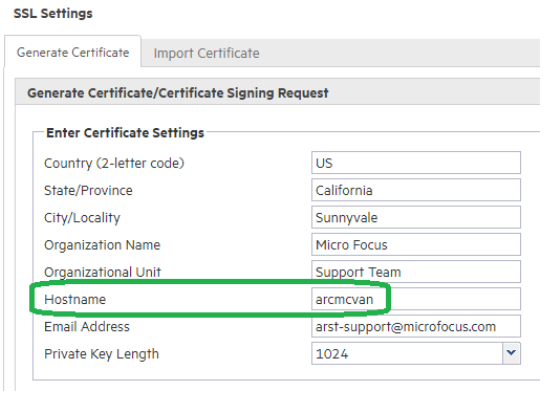
- Once Event Broker has been deployed, shutdown and take a snapshot
  - With EB 2.21, default “shutdown –h” command is sufficient to shut down EB instance (all required “clean” shutdown commands for EB have been rolled into the default shutdown process)

Pre-Req: Staging ArcMC to Manage EB

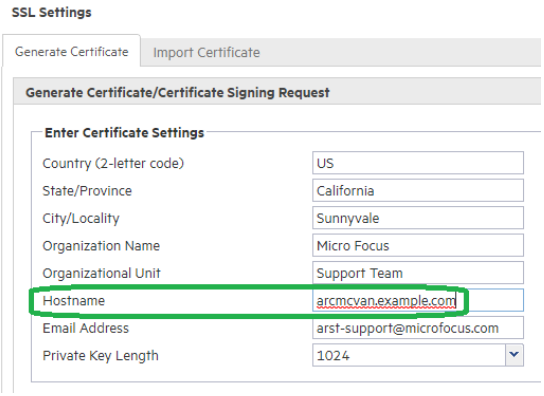
In this section, a new self-signed cert is generated on ArcMC, and then this new cert information is manually copied to the ArcSight Installer (for use with EB).

- Logon to ArcMC
- Click Administration -> System Admin -> SSL Server Certificate
- Update Hostname field to match FQDN for ArcMC

NOTE: By default in the certgen steps in ArcMC, *the hostname of the server (not the FQDN) is pre-populated in the UI*. *Be sure to update the hostname entry to FQDN before generating the new cert.*



update to ->



- Click Generate Certificate
  - When the cert is being re-generated and installed, the ArcMC UI will lose connectivity until everything refreshes
- Click Administration -> System Admin -> SSL Server Certificate -> View Certificate
- Copy the Server Certificate, this will be pasted in the ArcSight Installer interface
- Log on to the ArcSight Installer interface

  https://eb1.example.com:5443/

- Click Configuration > ArcSight Event Broker
- Update ArcMC hostname to FQDN: arcmcvan.example.com
- Replace text in ArcMC certificate field with pasted cert
- Click Save
  - Following should display at top of window:

Managing EB in ArcMC

In this section, the EB certificate is manually copied to ArcMC to enable ArcMC to communicate with the EB cluster.

- Start EB, and ensure all pods are running

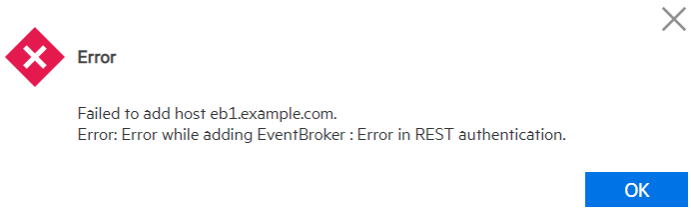
  ```
  watch -n 15 kubectl get pods --all-namespaces
  ```

- Follow steps to add EB node in ArcMC:
  - Logon to ArcMC UI
  - Click Node Management -> View All Nodes
  - Click Default
  - Click Add Host
    - Hostname/IP: `eb1.example.com`
    - Type: Event Broker 2.02 or later
    - Port: 38080 <leave default>
    - Cluster Port: 5443 <leave default>
    - Cluster Username: admin
    - Cluster Password: <value set on EB install>
  - Cluster Certificate:
    - On EB server, run following:

      ```
      /opt/arcsight/kubernetes/scripts/arcsight-cert-util.sh
      ```

    - Copy and paste the result to the Cluster Certificate field
  - Click Add

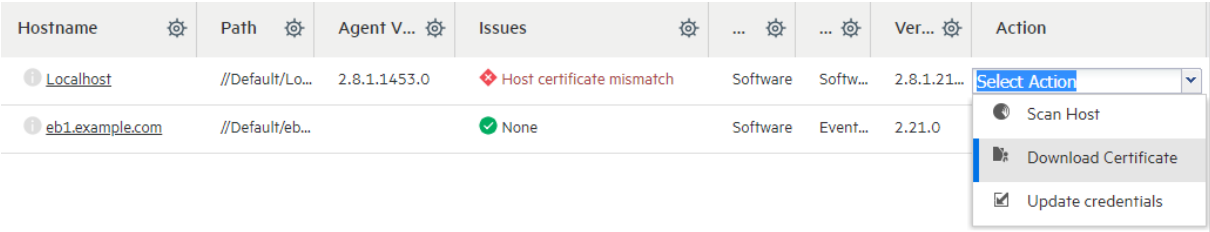
  - Click Continue
    - If newly generated ArcMC cert hasn't been added to the ArcSight Installer (or the EB cert script isn't used to get cert details for Add Host step), following error can display.



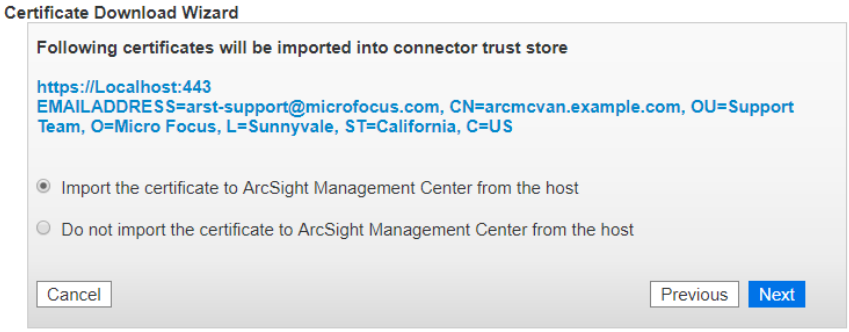
- From ArcMC UI, verify EB displays in Dashboard > Topology View

Updating ArcMC Agent on ArcMC

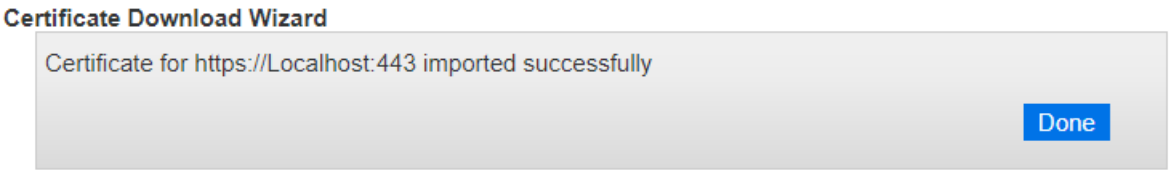
- Click Node Management > View All Nodes
- Click Default
    - Note the Host certificate mismatch
    - Click Download Certificate from the Action drop-down



    - Click Next
    - Click Next on the cert window



    - Click Done



Shutdown All Servers and Take Snapshots

At this stage, taking snapshot across all 3 servers (DNS, ArcMC, and EB1) to have a “known good” snapshot of the servers taken at the same time (even though no changes made to DNS)

- Shutdown eb1.example.com (then take snapshot)
    - With EB 2.21, prior needed commands (like sync, etc.) have been rolled into the default shutdown process

    ```
    shutdown -h now
    ```

- Shutdown dns.example.com (then take snapshot)
- Shutdown arcmcvan.example.com (then take snapshot)

Configuring and Including Tech Enablement Logger VM for 172.16.*

Start All Servers

- DNS
  - Wait for logon prompt
- ArcMC
  - Logon to ArcMC UI when bootup complete
- EB (wait for all EB pods to get to "running" status)
  - As root, from ssh window, track with following command once EB processes boot up enough to allow connection:

    `watch -n 15 kubectl get pods --all-namespaces -o wide`

  - Note: Even after all pods show as running, it will take a few minutes for ArcMC to update the status of EB to "green"
- Logger
  - Wait until able to logon to UI before proceeding to next steps

Logger 6.61 (as of 9/6/18)

Default settings for the Tech Enablement VM

```
Logger 6.61 demo VM

(Set to RAM=12G, Processors=4)
(Set to NAT and Bridged)

CentOS 7.3
Hostname: logger.arcsight.example.com
eth0 NAT: 172.16.100.100/24
eth1 Bridged: DHCP (off)
Default gateway: 172.16.100.2
OS users: root/arcsight, arcsight/arcsight
X display libraries installed
32-bit display libraries installed
ArcMC agent installed
Logger 6.61 installed to /opt/arcsight/logger

Browse to https://172.16.100.100 for Logger web UI
Logger user: admin/password

NOTE: If you don't have a Reports menu item in the Logger web UI, this VM
is running with a limited instant-on license, and a full license will
need to be applied for full functionality.
```

Configuration Changes to Local Laptop

- Added updated Logger entry to local system hosts file
  - For example: 172.16.100.100 logger.example.com

Verify Configuration Changes to DNS (BIND) server VM

The following changes were made based on the MacGyver Guide EB Companion 1 doc (if the EB companion doc used, some of these changes already made and just need to be verified):

- Logon as root to dns.example.com
- Change to /etc/named/zones
- If not already done (from MacGyver Companion Guide):
    - Edit example.com:
        - Update logger entry to 172.16.100.100 (from .65)
    - Edit 100.16.172.in-addr.arpa file:
        - Update 65 to 100 (last octet of IP)
- Update /etc/hosts:
    - Update entry for logger:

```
172.16.100.100   logger.example.com
```

Verify Configuration Changes to Event Broker (single master/worker node – eb1.example.com)

- Update /etc/hosts:
    - Add entry for logger:
        - 172.16.100.100 logger.example.com
- Verify nslookup (from eb1) for logger permutations:
    - logger
    - logger.example.com
    - 172.16.100.100

Verify Config Changes on ArcMC 2.81 VM for Updated Logger 6.61

- Power on ArcMC VM
- Log on as root
- Update /etc/hosts:
    - Add entry for logger:

```
172.16.100.100   logger.example.com
```

    - Add entries for DNS, EB1, and arcmcvan if not already listed

```
172.16.100.191   eb1.example.com
172.16.100.53 dns.example.com
172.16.100.242   arcmcvan.example.com
```

- Verify nslookup (from arcmcvan2) for logger permutations returns values from the 172.16.100.53 DNS server
    - logger
    - logger.example.com
    - 172.16.100.100

Configuration Changes to Logger 6.61 VM

- Log on as root
- Update /etc/hosts
    - Comment out any existing entries (legacy Logger, router, etc.)
    - Update hostname to “logger.example.com”
        - 172.16.100.100 logger.example.com

- Add entries for dns, arcmcvan, and eb1:

```
172.16.100.53    dns.example.com
172.16.100.242   arcmcvan.example.com
172.16.100.191   eb1.example.com
```

- Use nmtui:
  - Set system hostname to "logger.example.com"
  - Edit a connection - for eth0:
    - Set Gateway to: 172.16.100.2
    - Update DNS server to point to:
      - Primary: 172.16.100.53
      - Secondary: 172.16.100.2
    - Set search domain to: example.com

- Reboot Logger VM to completely restart networking (and take new DNS setting into account)

```
reboot
```

- Re-log on as root
- Verify networking is updated with DNS:

```
nslookup eb1
nslookup arcmcvan
```

- Verify perl installed by typing:
  - Perl is a pre-req for ArcMC agent installation
  - Step both verifies perl install (and updated to latest version), and also verifies the logger server can access external networks

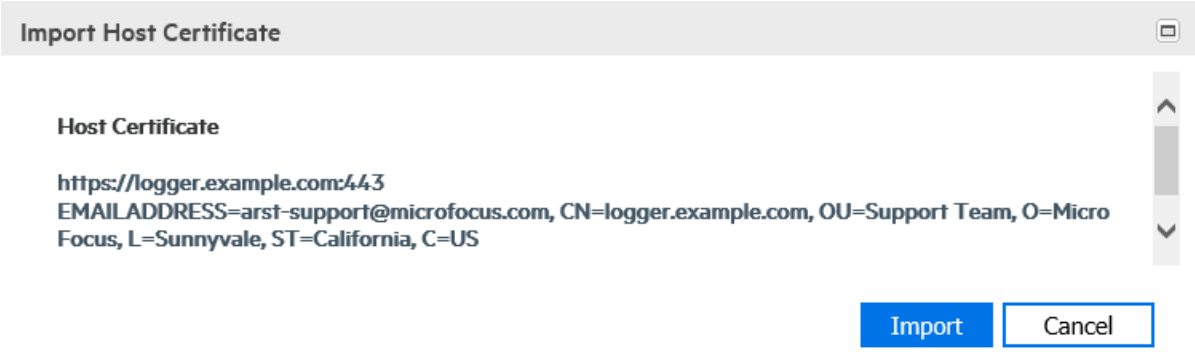
```
yum install perl
```

- Verify ability to logon to Logger
  - https://logger.example.com
    - If UI doesn't display, verify entries in local hosts file (and ensure logger.example.com first in row of aliases)
  - admin / password

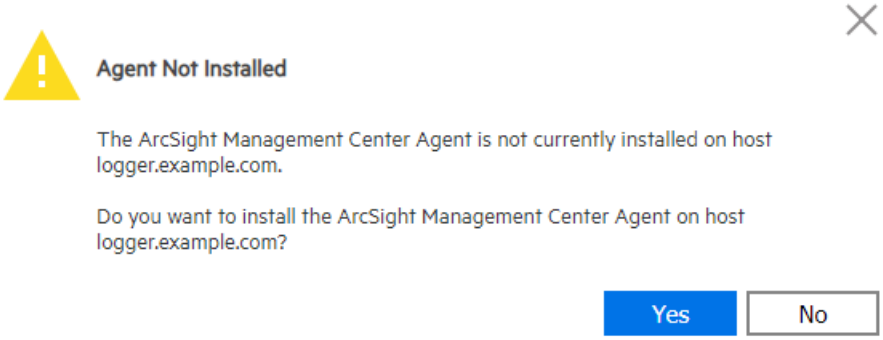
- From Logger UI, update Logger cert
  - System Admin > SSL Server Certificate
  - Verify hostname shows as: logger.example.com
    - The hostname will populate with the hostname defined on the server
  - Click Generate Certificate
    - Navigation to webpage will show as cancelled temporarily while new cert generated and applied
  - Re-log on to UI, to verify connectivity
  - Log off Logger UI

Add Logger 6.6.1 as Managed Node in ArcMC 2.81 VM

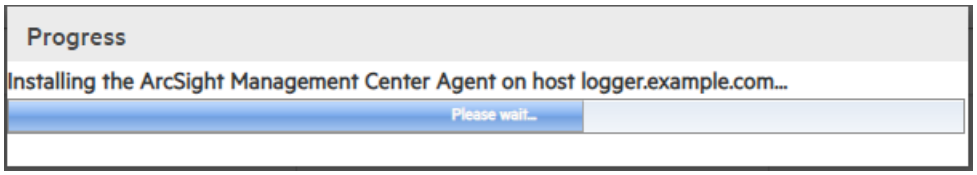
- Log on to ArcMC UI
- Click Node Management > View All Nodes
- Click on Default (if new ArcMC install)
- Click Add Host
- For hostname/IP, use: `logger.example.com`
- For type, use: `Software Form Factor`
- Host credentials: Logger admin user / pw
- Port: `443`
- Click Add
    - When prompted, click Import to import host certificate



    - When prompted, click Yes to install ArcMC agent



Agent is installed:



Once added, entry displays in list:

Hostname/IP	Type	Date Added
logger.example.com	Software Logger	Sep 7, 2018, 10:36:37 AM

- Click Dashboard > Monitoring Summary
- Click Loggers
    - Verify status of the added Logger

Logger Severity Issue Summary

Name	Path	Type	Lead/Breach
logger.example.com	//ArcNet/logger.example.com/Software Logger	Software Logger	Healthy

- Click Dashboard > Topology View
    - Verify a Logger destination is displayed

Destinations

logger.example.com

- Click on the logger.example.com destination, and wait for status to display in ArcMC

Hardware Status

Search All

Name	Type	Time Stamp	Status
Disk/Space/Remaining/R...	DISK_SPACE	Sep 24, 2018, 09:51:10 ...	Ok
CustomerA	STORAGE_GROUP_USAGE	Sep 24, 2018, 09:51:13 ...	1 GB out of 3 GB : 33....
CustomerB	STORAGE_GROUP_USAGE	Sep 24, 2018, 09:51:13 ...	1 GB out of 3 GB : 33....
CustomerC	STORAGE_GROUP_USAGE	Sep 24, 2018, 09:51:13 ...	1 GB out of 3 GB : 33....
CustomerD	STORAGE_GROUP_USAGE	Sep 24, 2018, 09:51:13 ...	1 GB out of 3 GB : 33....
Default Storage Group	STORAGE_GROUP_USAGE	Sep 24, 2018, 09:51:13 ...	1 GB out of 4 GB : 25....
Flow data	STORAGE_GROUP_USAGE	Sep 24, 2018, 09:51:13 ...	1 GB out of 4 GB : 25....
Internal Event Storage Gr...	STORAGE_GROUP_USAGE	Sep 24, 2018, 09:51:13 ...	1 GB out of 3 GB : 33....

Shutdown All Servers and Take Snapshots

At this stage, taking snapshot across all 4 servers (DNS, ArcMC, EB1, and Logger) to have a “known good” snapshot of the servers taken at the same time (even though no changes made to DNS at this stage)

- Shutdown eb1.example.com (then take snapshot)
    - With EB 2.21, prior needed commands (like sync, etc.) have been rolled into the default shutdown process

    ```
    shutdown -h now
    ```

- Shutdown dns.example.com (then take snapshot)
- Shutdown arcmcvan.example.com (then take snapshot)
- Shutdown logger.example.com (then take snapshot)

Appendix A:BIND Supporting Files

named.conf (/etc/named.conf)

```
options {
listen-on port 53 { any; };
# listen-on-v6 port 53 { any; };
forwarders { 172.16.100.254; };
directory "/var/named";
dump-file "/var/named/data/cache_dump.db";
statistics-file "/var/named/data/named_stats.txt";
memstatistics-file "/var/named/data/named_mem_stats.txt";
allow-query { any; };
allow-transfer { none; };
recursion yes;
dnssec-enable no;
dnssec-validation no;
dnssec-lookaside auto;
/* Path to ISC DLV key */
bindkeys-file "/etc/named.iscdlv.key";
managed-keys-directory "/var/named/dynamic";
};
logging {
channel default_debug {
file "data/named.run";
severity dynamic;
};
};
zone "." IN {
type hint;
file "named.ca";
};
include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";
include "/etc/named/named.conf.local";
```

named.conf.local (/etc/named/named.conf.local)

```
zone "example.com" IN {
type master;
file "/etc/named/zones/example.com";
allow-transfer { 172.16.100.254; };
};
zone "100.16.172.in-addr.arpa" IN {
type master;
file "/etc/named/zones/100.16.172.in-addr.arpa";
allow-transfer { 172.16.100.254; };
};
```

example.com (/etc/named/zones/example.com)

```
$TTL 3H
@ IN SOA @ hostmaster.example.com. (
0 ; serial
3H ; refresh
1H ; retry
1W ; expire
3H ) ; minimum
@ IN NS dns.example.com.
@ IN A 172.16.100.53
dns IN A 172.16.100.53
eb1 IN A 172.16.100.191
eb2 IN A 172.16.100.192
eb3 IN A 172.16.100.193
eb4 IN A 172.16.100.194
eb5 IN A 172.16.100.195
eb6 IN A 172.16.100.196
eb IN A 172.16.100.22
vertica1 IN A 172.16.100.211
vertica2 IN A 172.16.100.212
vertica3 IN A 172.16.100.213
arcmc IN A 172.16.100.117
arcmcvan IN A 172.16.100.242
esm IN A 172.16.100.109
logger IN A 172.16.100.100
nfs IN A 172.16.100.111
```

2.4 100.16.172.in-addr.arpa (/etc/named/zones/100.16.172.in-addr.arpa)

```
$TTL 3H
@ IN SOA @ hostmaster.example.com. (
0 ; serial
3H ; refresh
1H ; retry
1W ; expire
3H ) ; minimum
@ IN NS dns.example.com.
53 IN PTR dns.example.com.
191 IN PTR eb1.example.com.
192 IN PTR eb2.example.com.
193 IN PTR eb3.example.com.
194 IN PTR eb4.example.com.
195 IN PTR eb5.example.com.
196 IN PTR eb6.example.com.
22 IN PTR eb.example.com.
211 IN PTR vertica1.example.com.
212 IN PTR vertica2.example.com.
213 IN PTR vertica3.example.com.
117 IN PTR arcmc.example.com.
242 IN PTR arcmcvan.example.com.
109 IN PTR esm.example.com.
100 IN PTR logger.example.com.
111 IN PTR nfs.example.com.
```

Appendix B: Misc Notes

Disabling Default Firewall <optional>

With CentOS, firewalld is enabled by default. Depending on the product to be installed on the base OS, this OS may need to remain on (for example, EB), or can be disabled if not used.
The following steps outline quick example for disabling FW if wanted / needed.

1. On DNS server, logon as root
2. Type following to check FW status:
   `firewall-cmd --state`
   a. Should show: "running"
3. Type following to stop firewall
   `systemctl stop firewalld`
4. Type following to disable firewall:
   `systemctl disable firewalld`
5. Re-check firewall status:
   `firewall-cmd –state`
   a. Should show:"not running"

Misc Port-checking Commands

- netstat
  - netstat (network statistics) is a command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables, interface statistics etc. netstat is one of the most basic network service debugging tools, telling you what ports are open and whether any programs are listening on ports.
    - netstat comes with the net-tools package installation

  - Listing all the LISTENING Ports of TCP and UDP connections
    netstat –a

  - Listing all LISTENING Connections
    netstat –l

  - Listing all UNIX Listening Ports
    netstat -lx

  - Display all active connections:
    netstat –a (display all active connections)
    netstat –an (display all active connections by IP address vs. hostname)

- ss
  - To see if a program or process is listening on a port, ready to accept a packet, use the ss program.

```
# ss -nutlp
The arguments to the ss program are listed below:

t – Display TCP sockets.
u – Display UDP sockets.
l – Display listening sockets
n – Do now try to resolve names
p – Show process using socket
```

Iptables Command

firewalld is a firewall management tool for Linux operating systems. It provides firewall features by acting as a front-end for the Linux kernel's netfilter framework via the iptables command, acting as an alternative to the iptables service.

With CentOS 6.4, the following command can be issued to see an overview of services, and enabled ports and protocols:

```
iptables -L -n
```

FirewallD Commands

- To start the service and enable FirewallD on boot:

```
systemctl start firewalld
systemctl enable firewalld
```

- To stop and disable it:

```
systemctl disable firewalld
```

- To check the firewall status. The output should say either running or not running.

```
firewall-cmd --state
```

```
[root@eb1 ~]# firewall-cmd --state
running
```

- To view the status of the FirewallD daemon:

systemctl status firewalld

```
[root@dns services]# systemctl status firewalld
* firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; vendor preset:
enabled)
   Active: active (running) since Wed 2018-08-22 11:13:01 EDT; 21min ago
     Docs: man:firewalld(1)
 Main PID: 759 (firewalld)
   CGroup: /system.slice/firewalld.service
           └─759 /usr/bin/python -Es /usr/sbin/firewalld --nofork --nopid

Aug 22 11:13:01 dns systemd[1]: Starting firewalld - dynamic firewall daemon...
Aug 22 11:13:01 dns systemd[1]: Started firewalld - dynamic firewall daemon.
Aug 22 11:13:02 dns firewalld[759]: WARNING: ICMP type 'beyond-scope' is no...6.
Aug 22 11:13:02 dns firewalld[759]: WARNING: beyond-scope: INVALID_ICMPTYPE...e.
Aug 22 11:13:02 dns firewalld[759]: WARNING: ICMP type 'failed-policy' is n...6.
Aug 22 11:13:02 dns firewalld[759]: WARNING: failed-policy: INVALID_ICMPTYP...e.
Aug 22 11:13:02 dns firewalld[759]: WARNING: ICMP type 'reject-route' is no...6.
Aug 22 11:13:02 dns firewalld[759]: WARNING: reject-route: INVALID_ICMPTYPE...e.
Hint: Some lines were ellipsized, use -l to show in full.
```

Additional Firewall-related Notes

FirewallD can allow traffic based on predefined rules for specific network services. You can create your own custom service rules and add them to any zone. The configuration files for the default supported services are located at /usr/lib/firewalld/services and user-created service files would be in /etc/firewalld/services.

To view the default available services:
sudo firewall-cmd --get-services

For DNS, can look at dns.xml

```
[root@dns services]# vi dns.xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>DNS</short>
  <description>The Domain Name System (DNS) is used to provide and request host and domain names.
Enable this option, if you plan to provide a domain name service (e.g. with bind).</description>
  <port protocol="tcp" port="53"/>
  <port protocol="udp" port="53"/>
</service>
~
"dns.xml" 7L, 346C
```

= = = = =

Public is the default zone set, if you do not change it. To check the currently set default zone use the below command:

firewall-cmd --get-default-zone

[root@dns ~]# firewall-cmd --get-default-zone
public

= = = = =

To list the ports that are open on your system:

firewall-cmd --list-ports

<By default, minimal CentOS install, no result displays>

= = = = =

To show currently allowed service on your system use the below command.

firewall-cmd --list-services

[root@dns ~]# firewall-cmd --list-services
ssh dhcpv6-client

= = = = =

Allowing or Denying an Arbitrary Port/ProtocolPermalink

As an example: Allow or disable TCP traffic on port 12345.

sudo firewall-cmd --zone=public --add-port=12345/tcp --permanent
sudo firewall-cmd --zone=public --remove-port=12345/tcp --permanent

= = = = =

To make changes permanent, use the –permanent option. Example:

firewall-cmd --permanent --zone=public --add-service=http

firewall-cmd --permanent --zone=public --add-service=dns

= = = = =

firewall-cmd --zone=public --list-services

Changing Server Hostname

1. Using a text editor, open the server's **/etc/sysconfig/network** file.

```
# sudo nano /etc/sysconfig/network
```

2. Modify the `HOSTNAME=` value to match your FQDN hostname.

```
HOSTNAME=myserver.domain.com
```

3. For internal networking, change the host that is associated with the main IP address for your server (found at **/etc/hosts**).

```
127.0.0.1       localhost localhost.localdomain
123.45.67.89   hostname.domain.com   hostname
```

4. Run the `hostname` command. This command lets you change the hostname on the server that the command line remembers, but it does not actively update all programs that are running under the old hostname.

```
[root@defiant ~]# hostname hostname.domain.com
[root@defiant ~]# hostname
hostname.domain.com
[root@defiant ~]#
```

5. Restart networking on your server to ensure that changes will persist on restart.

```
# /etc/init.d/network restart
```

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